Learning-Based Adaptive Control of a Hydraulic Manipulator

Control Architecture, Dataset Rationale, Experimental Plan, and Detailed Thesis Outline

PPO1: ID-FD Pipeline | PPO2: ROM-AAM Pipeline

Central thesis idea

A common upper-level residual PPO controller is evaluated with two learned hydraulic-actuator representations. PPO1 uses a learned inverse-dynamics/forward-dynamics pipeline as a non-adaptive baseline. PPO2 uses a reduced-order cylinder model and introduces online adaptation through an Arm Adaptation Module (AAM) that estimates latent dynamics from recent observations.

Prepared as a working plan for thesis implementation and evaluation

1. Thesis Scope and Main Research Question

The thesis investigates whether a hydraulic manipulator can achieve accurate and robust task-space tracking by combining: **a nominal Damped Least-Squares (DLS) controller, an upper-level residual PPO policy, learned hydraulic actuator dynamics, and online latent adaptation.**

Main research question

Does a latent-conditioned reduced-order hydraulic cylinder model, adapted online through an AAM, improve trajectory tracking and robustness under changing hydraulic and mechanical conditions compared with non-adaptive learned actuator models?

1.1 Role of each controller component

Component	Role	What is learned?
DLS controller	Provides the nominal task-space-to-joint/cylinder correction and a structured motion baseline.	No
Residual PPO	Learns high-level corrections that improve tracking and compensate for remaining model/controller errors.	Yes
ID-FD pipeline	Maps desired motion/state to valve commands and represents/predicts actuator response.	Yes
Reduced-order model (ROM)	Represents dominant cylinder-space hydraulic dynamics with lower complexity than the full Simscape model.	Partly/Yes
Arm Adaptation Module (AAM)	Infers a latent representation of the current arm and actuator dynamics from recent observation history.	Yes

1.2 Why the title remains appropriate

The title ***Learning-Based Adaptive Control of a Hydraulic Manipulator*** is supported because the complete controller is learning-based and the PPO2 method contains an explicit online adaptation mechanism. The AAM estimates latent dynamics from observations and conditions the ROM

according to the currently experienced system. DLS being classical does not weaken the title; it makes the architecture hybrid and structured rather than fully black-box.

2. Why the Simscape Dataset Was Randomized

The randomization performed during Simscape data generation is fundamental to both robustness and adaptation. A dataset collected only from one nominal parameter set would teach the models one fixed hydraulic system. It would not contain enough variation to learn how actuator behaviour changes across operating conditions.

2.1 Purposes of domain randomization

- Expose the learned models to variations in hydraulic parameters, valve characteristics, compressibility, leakage, friction, mechanical properties, payload-related effects, and other uncertain dynamics represented in the Simscape model.
- Prevent the learned ROM and other models from overfitting to a single nominal plant.
- Create a family of related dynamical systems from which a compact latent representation can be learned.
- Provide supervised context for training the teacher latent encoder or system-dynamics representation used to define the AAM target.
- Enable the AAM to learn the inverse problem: infer which type of dynamics is currently active from a recent history of states, pressures, motions, and commands.
- Support evaluation on held-out and unseen parameter combinations, which is necessary to demonstrate robustness and adaptation rather than memorization.

> **Important distinction**
> Randomized training alone produces robustness: one model performs reasonably across many systems. AAM conditioning produces adaptation: the model changes its response according to an estimated latent description of the current system.

2.2 Why both closed-loop and open-loop experiments were collected

Dataset source	Main strength	Main limitation	Use in this thesis
Closed-loop experiments	Realistic trajectories, controller-regulated behaviour, safe operating regions, and state distributions similar to the final task.	The feedback controller can correlate commands and motion, suppress excitation, and hide some valve-to-cylinder dynamics.	Provides realistic task-space and tracking behaviour; useful for policy-relevant distributions and stable operating regimes.
Open-loop experiments	Direct excitation of valve-to-pressure-to-cylinder dynamics; stronger independent input variation and	May include motions not commonly visited by the final controller and requires careful safety/constraint	Improves identification of causal hydraulic response, transient behaviour, delay, directionality,

Dataset source	Main strength	Main limitation	Use in this thesis
	better actuator identifiability.	design.	gain, pressure dependence, and saturation.
Combined dataset	Balances realism and identifiability.	Requires balanced sampling and split-by-run validation to avoid leakage.	Used to train or fine-tune ROM/ID/FD models so they remain useful in closed-loop control but still respond correctly to valve excitation.

3. Overall Hierarchical Control Architecture

1. The desired end-effector trajectory defines the task-space reference.
2. The DLS controller converts task-space error into a nominal joint or cylinder-space motion correction.
3. The residual PPO policy observes tracking error, robot state, actuator state, and relevant history and produces an upper-level correction.
4. The selected hydraulic pipeline converts the combined motion intent into cylinder/valve behaviour inside IsaacLab.
5. The same trajectory families, reward definitions, evaluation metrics, simulation settings, and comparable training budgets should be used across methods.

Experimental control principle
PPO is present in every principal comparison. Therefore, the experimental variable is the hydraulic actuator representation and its adaptation mechanism - not whether PPO is used.

4. PPO1 - ID-FD Pipeline

PPO1 is the learned inverse-dynamics/forward-dynamics baseline. It does not use the AAM. Its purpose is to establish how well a non-adaptive learned actuator pipeline can support the same upper-level residual PPO tracking task.

4.1 PPO1-A: Nominal ID-FD

- ID and FD are trained primarily for the nominal system or nominal parameter region.
- No online latent estimation and no AAM conditioning.
- Evaluated first on nominal parameters, then under parameter shifts to quantify sensitivity.
- Provides the simplest learned-model baseline.

4.2 PPO1-B: Randomized/robust ID-FD

- ID and/or FD are trained or fine-tuned using the randomized Simscape dataset, including useful open-loop excitation data.
- No AAM is applied to the FD pipeline.
- The model learns one shared mapping that must work across the randomized family of systems.
- This tests whether domain-randomized model training alone is enough for robust control.

Interpretation of PPO1
PPO1-B is robust but not explicitly adaptive. It has seen parameter variation during training, but it does not estimate a system-specific latent state and switch its internal dynamics online.

5. PPO2 - Reduced-Order Model Pipeline

PPO2 replaces the separated ID-FD actuator representation with a hydraulic-aware reduced-order cylinder model. The ROM should preserve the dominant valve, pressure, cylinder-position, and

cylinder-velocity relationships needed by the IsaacLab plant while remaining compact enough for fast control and reinforcement-learning rollouts.

5.1 PPO2-A: Nominal ROM

- ROM trained for nominal dynamics or evaluated with a fixed nominal parameterization.
- No AAM and no latent conditioning.
- Used to compare the ROM architecture fairly against PPO1-A without adaptation.

5.2 PPO2-B: Randomized/robust ROM

- ROM trained across the randomized Simscape systems using closed-loop and open-loop data.
- No online AAM; the ROM receives no estimated system-specific latent vector.
- Tests the direct benefit of randomized training on ROM robustness.

5.3 PPO2-C: AAM-conditioned adaptive ROM - proposed method

- The AAM receives recent observation history, such as cylinder motion, pressure information, joint/arm state, previous commands, and other selected measurable signals.
- The AAM estimates a compact latent vector representing the currently experienced hydraulic and mechanical dynamics.
- The ROM is conditioned on this estimated latent vector so its predicted transition or actuator response adapts online.
- The residual PPO remains the same upper-level controller role used in the other experiments.
- This configuration is the main proposed adaptive contribution of the thesis.

5.4 Optional diagnostic: Oracle-latent ROM

During simulation-only evaluation, the ROM can optionally receive the true context or teacher latent generated from the known randomized parameters. This is not a deployable controller. It is an upper-bound diagnostic that separates ROM capacity from AAM estimation error.

6. Final Experiment Matrix

Experiment	PPO	Actuator model	Randomized training	Latent input	AAM online	Purpose
PPO1-A	Yes	Nominal ID-FD	No/nominal	No	No	Nominal learned baseline
PPO1-B	Yes	Robust ID-FD	Yes	No	No	Effect of randomization on ID-FD
PPO2-A	Yes	Nominal ROM	No/nominal	No	No	Fair ROM architecture baseline
PPO2-B	Yes	Robust ROM	Yes	No	No	Effect of randomization on ROM
PPO2-C	Yes	Latent-conditioned ROM	Yes	Estimated latent	Yes	Main adaptive method
PPO2-D, optional	Yes	Latent-conditioned ROM	Yes	Oracle/ teacher latent	No estimator	Upper-bound diagnostic

6.1 Required fairness controls

- Same task-space trajectories and target poses.
- Same DLS formulation, limits, control frequency, simulation step, and action scaling.
- Same residual PPO observation/action definition wherever technically possible.
- Same reward terms and termination conditions.
- Comparable PPO network size, training steps, seeds, and checkpoint-selection rule.
- Run-level train/validation/test splits to prevent samples from the same Simscape run leaking across splits.
- Held-out parameter combinations for the final robustness/adaptation test.

7. Evaluation Protocol

7.1 Reference trajectories

Use the same trajectory set for all methods: straight lines, circular or semicircular arcs, smooth curves, and repeated mowing-style sweeps. Include stationary-truck arm tracking and, where required by the thesis scenario, movement to a new station followed by a new tracking task.

7.2 Test conditions

- Nominal plant parameters.
- Single-factor hydraulic variations.
- Single-factor mechanical or payload-related variations.
- Combined randomized conditions within the training distribution.
- Held-out combinations and extrapolation conditions outside the exact training samples.
- Abrupt or gradual parameter changes within an episode, when feasible, to measure adaptation time.

7.3 Metrics

Metric group	Recommended measures
Task tracking	End-effector RMSE, MAE, maximum error, path error, orientation error where applicable.
Control quality	Residual PPO action magnitude, action variance, smoothness, valve saturation rate, command-rate violations.
Model quality	One-step error, multi-step rollout error, pressure/cylinder transition error, direction and gain consistency.
Robustness	Mean and worst-case performance across randomized seeds and held-out parameter sets.
Adaptation	Latent-estimation error where teacher targets exist, adaptation settling time, post-change tracking recovery.
Efficiency	Inference time, simulation throughput, memory use, and stability over long rollouts.

Critical interpretation rule

Because PPO can compensate for an imperfect actuator model, tracking error alone is insufficient. A better actuator representation should also require smaller, smoother residual actions and should show lower independent model-rollout error.

8. Thesis Hypotheses and Expected Claims

- H1: Randomized ID-FD training improves robustness over nominal ID-FD under hydraulic and mechanical parameter changes.
- H2: A randomized ROM improves robustness over a nominal ROM.

- H3: The AAM-conditioned ROM outperforms the same randomized ROM without AAM on held-out and changing dynamics.
- H4: The adaptive ROM reduces the magnitude and variability of the PPO residual correction required for the same tracking task.
- H5: Open-loop excitation data improves actuator identifiability and ROM/FD dynamic fidelity compared with relying only on closed-loop data.
- H6: The full PPO2-C method approaches the oracle-latent upper bound closely enough to demonstrate useful online latent estimation.

8.1 Claims that should remain separate

- ID-FD versus fixed ROM compares two learned actuator-model architectures.
- Nominal versus randomized training measures robustness gained from dataset diversity.
- Randomized ROM versus AAM-conditioned ROM isolates the contribution of online adaptation.
- Best ID-FD versus adaptive ROM compares the strongest non-adaptive baseline with the complete proposed method.

9. Recommended Novelty Statement

Proposed thesis contribution

This thesis develops and evaluates a latent-conditioned reduced-order model of hydraulic cylinder dynamics for a mobile manipulator. The model is trained from domain-randomized high-fidelity Simscape data combining realistic closed-loop trajectories with informative open-loop actuator excitation. An Arm Adaptation Module infers a compact latent representation of the current system dynamics from recent observations and conditions the ROM online. The adaptive ROM is integrated with a DLS nominal controller and a residual PPO policy and is compared against nominal and randomized ID-FD and ROM baselines under common task-space tracking experiments.

10. Recommended Thesis Title

Learning-Based Adaptive Control of a Hydraulic Manipulator

More explicit alternative:

Learning-Based Adaptive Control of a Hydraulic Manipulator Using Latent-Conditioned Reduced-Order Dynamics

11. Immediate Implementation Checklist

- Finalize the exact input/output contract of the cylinder-space ROM.
- Define how the randomized context or teacher latent is produced for every Simscape run.
- Train the randomized ROM with balanced closed-loop and open-loop samples.
- Train the AAM to estimate the teacher latent from deployable observation histories only.
- Verify that the ROM changes meaningfully when the latent vector changes.
- Integrate PPO2-B first, then add AAM conditioning to form PPO2-C.
- Freeze the final experiment matrix, metrics, seeds, and held-out parameter sets before long PPO training.

- Report both control-level and model-level metrics so PPO compensation does not hide actuator-model weaknesses.

PART II

Recommended Standard Thesis Structure

Eight main chapters following a conventional engineering-thesis progression

Structural principle used
The chapter names are intentionally conventional. The hydraulic manipulator remains the technical subject of the thesis, but the word “hydraulic” does not need to appear in every chapter title. The structure follows the logic of the provided RPTU dissertation: introduction, conceptual foundations, system fundamentals, focused technical chapters, integrated evaluation, and conclusion.

12. How the Sample Thesis Is Adapted

The sample dissertation does not force the application domain into every chapter title. Instead, it uses standard titles such as Introduction, System Fundamentals, Decision Making, and Conclusion, while keeping the application visible through the content of each chapter. The same principle is appropriate here.

Logic in the sample	Adaptation for this thesis	Purpose
Introduction	Introduction	Motivation, problem, questions, contributions, scope, and thesis map.
Core principles	Fundamentals and Related Work	Establishes the theoretical basis and identifies the research gap.
System fundamentals	System Fundamentals and Problem Formulation	Defines the manipulator, simulations, coordinates, interfaces, and control problem.
Technical component chapters	Data Generation; Learned Models; Control Framework	Develops the thesis methods in a logical sequence.
Integrated framework and evaluation	Experimental Evaluation and Discussion	Combines all components and tests the claims.
Conclusion	Conclusion and Outlook	Answers the research questions and identifies limitations and future work.

12.1 Recommended narrative flow

- Why the control problem matters and what is missing in current methods.
- What technical principles are required to understand the proposed approach.
- What system is controlled and how the control problem is formulated.

- How the randomized Simscape datasets were produced and why both closed-loop and open-loop excitation were required.
- How the ID, FD, ROM, latent representation, and AAM were developed.
- How DLS, residual PPO, and the learned models were integrated into PPO1 and PPO2.
- Whether the proposed adaptive method improves prediction, tracking, robustness, and control effort.
- What can be concluded, what remains limited, and what should be investigated next.

Central writing rule
Each chapter should advance one part of the research argument. Do not repeat a generic description of hydraulic systems in every chapter. Explain the physical system once in depth, then refer back to the relevant variables, uncertainties, and actuator effects when they are needed for the model, controller, or experiment.

13. Recommended Front Matter

- **Title page:** Official title, author, degree programme, institute, supervisors, and submission date.
- **Declaration:** Academic-integrity statement required by the university.
- **Abstract:** Problem, approach, datasets, PPO1/PPO2 architecture, principal results, and contribution in approximately 250-400 words.
- **Acknowledgements:** Brief acknowledgements, if desired.
- **Contents and lists:** Table of contents, list of figures, list of tables, and list of abbreviations or symbols.
- **Nomenclature:** Define q, q-dot, cylinder extension, cylinder velocity, differential pressure, valve command, latent vector, PPO action, and major model abbreviations.

Chapter 1: Introduction

Introduce the practical and scientific problem, motivate the proposed learning-based adaptive architecture, and state exactly what the thesis contributes.

Chapter purpose	Main output	Indicative length
Introduce the practical and scientific problem, motivate the proposed learning-based adaptive architecture, and state exactly what the thesis contributes.	A clear research problem, research questions, objectives, contributions, scope, and chapter roadmap.	6-9 pages

1.1 Motivation and application context

- Introduce heavy-duty and mobile manipulators and explain why hydraulic actuation is used when high force density and robustness are required.
- Explain the difficulty of precise end-effector tracking when the actuator response is nonlinear, load dependent, uncertain, and not represented adequately by a purely kinematic controller.
- Motivate a structured approach that retains a nominal task-space controller while learning the actuator behaviour and residual corrections.

1.2 Problem statement

- State the gap between task-space motion commands and the valve/cylinder behaviour required to realize them.
- Explain why a fixed nominal model can fail when hydraulic or mechanical parameters vary.
- Define the need for a model that is compact enough for reinforcement-learning simulation but expressive enough to represent actuator variability.

1.3 Research questions

- RQ1: Can learned cylinder-space dynamics improve task-space tracking when combined with a nominal DLS controller and residual PPO?
- RQ2: Does training on randomized Simscape data improve robustness compared with nominal-only model training?
- RQ3: Does an AAM-estimated latent representation improve the ROM beyond robust training without online adaptation?
- RQ4: How does the proposed ROM-AAM pipeline compare with the learned ID-FD pipeline under matched control and test conditions?

1.4 Objectives

- Generate informative nominal and randomized datasets from a high-fidelity Simscape model.
- Train and validate cylinder-space ID, FD, and reduced-order models.
- Learn a latent representation of randomized system dynamics and estimate it online with an AAM.
- Integrate PPO1 and PPO2 pipelines into IsaacLab and evaluate tracking, robustness, adaptation, and computational behaviour.

1.5 Contributions

- A combined closed-loop and open-loop data-generation strategy for learning cylinder-space actuator dynamics.
- A learned ID-FD control pipeline used as a non-adaptive learning-based reference architecture.
- A latent-conditioned reduced-order model and AAM-based online adaptation mechanism.
- A residual PPO task-space control framework integrating classical kinematics with learned actuator models.
- A controlled comparison across nominal, randomized/robust, and adaptive configurations.

1.6 Scope and limitations

- State whether the work is simulation-only or includes any real-system validation.
- Define which arm degrees of freedom and cylinders are modeled, including the passive coupling of the fifth joint.
- Clarify that the latent representation is not automatically equivalent to identifiable physical parameters.
- State which safety, hardware, and real-time deployment questions are outside the thesis scope.

1.7 Thesis structure

- Summarize Chapters 2-8 in one concise paragraph each, following the final structure below.

Chapter 2: Fundamentals and Related Work

Provide only the theory and literature needed to understand the research choices, then end with the unresolved gap addressed by the thesis.

Chapter purpose	Main output	Indicative length
Provide only the theory and literature needed to understand the research choices, then end with the unresolved gap addressed by the thesis.	A coherent conceptual foundation and a precise positioning of the thesis against prior work.	12-18 pages

2.1 Manipulator actuation and control fundamentals

- Electro-hydraulic actuation chain: valve command, flow, chamber pressures, cylinder force, cylinder motion, joint motion, and end-effector motion.
- Relevant nonlinearities and uncertainties: compressibility, entrained air, leakage, valve dead zone, saturation, friction, pressure limits, load dependency, and coupling.
- Why high-fidelity models are useful for data generation but expensive or inconvenient for large-scale RL training.

2.2 Kinematics and task-space tracking

- Forward kinematics and manipulator Jacobian.
- Inverse differential kinematics and Damped Least-Squares near singular configurations.
- Cylinder-to-joint and joint-to-cylinder coordinate mappings, including limitations introduced by coupling and passive joints.

2.3 Model-based and conventional control approaches

- Position, velocity, pressure, force, and cascade-control structures used for hydraulic manipulators.
- Strengths and limitations of PD, computed-torque, backstepping, adaptive, and robust methods in the context of uncertain hydraulic dynamics.
- The role of structured nominal controllers as a safe and interpretable baseline.

2.4 Data-driven system identification and learned dynamics

- Inverse dynamics, forward dynamics, one-step prediction, and multi-step rollout concepts.
- Temporal networks and history windows for dynamic systems.
- Input identifiability and the danger of learning kinematic persistence instead of control authority from closed-loop data.

2.5 Reduced-order modeling

- Purpose of reducing a high-fidelity model to dominant state transitions suitable for repeated simulation and control.
- Trade-off between physical interpretability, predictive accuracy, stability, and computational efficiency.
- Learned, grey-box, and latent-conditioned ROM variants relevant to this work.

2.6 Domain randomization and latent adaptation

- Domain randomization as exposure to families of hydraulic and mechanical systems rather than one nominal plant.
- Robust policies/models versus explicitly adaptive models.
- Teacher latent representations, history-based adaptation modules, and online system-context estimation.

2.7 Reinforcement learning and residual control

- PPO fundamentals required for the thesis: policy, value function, clipped objective, observations, actions, and reward shaping.
- Residual RL as correction of a nominal controller rather than complete replacement.
- Benefits and risks of allowing PPO to compensate for model error.

2.8 Research gap and thesis positioning

- Identify what existing work does not jointly provide: cylinder-space learned dynamics, randomized high-fidelity data, latent ROM adaptation, and residual task-space control within one evaluated framework.
- State how PPO1, PPO2, and the AAM ablation directly address the gap.

Chapter 3: System Fundamentals and Problem Formulation

Describe the physical and simulated system once, define all coordinate spaces and interfaces, and formulate the control objective mathematically.

Chapter purpose	Main output	Indicative length
Describe the physical and simulated system once, define all coordinate spaces and interfaces, and formulate the control objective mathematically.	A reproducible system definition and a precise contract between DLS, PPO, learned models, and the simulator.	10-15 pages

3.1 Manipulator platform

- Present the Mulag FME500 arm, mounting platform, relevant links, joints, cylinders, ranges of motion, and end effector.
- Explain the four actively modeled cylinders and the passive/coupled fifth joint.
- Provide a figure showing the arm, joint axes, cylinder locations, and coordinate frames.

3.2 Kinematic structure and coordinate mappings

- Define joint position and velocity vectors.
- Define cylinder extension and extension-velocity vectors.
- Describe the mapping between cylinder and joint motion and how signs, scales, offsets, and coupling are validated.
- Define the end-effector pose and task-space error used by DLS and PPO.

3.3 Actuation and measured state

- Define valve commands, differential pressure, optional force signals, and previous-action histories.
- Explain which quantities are available in Simscape, IsaacLab, and the proposed controller.
- Separate commands, measured observations, estimated variables, and learned latent variables.

3.4 High-fidelity Simscape model

- Describe the physical subsystems represented in Simscape and the simulation time step.
- Explain the model's role as a data generator and source of randomized dynamics.
- List the key assumptions and limitations of the high-fidelity model.

3.5 IsaacLab environment

- Describe the robot asset, cylinder actuation abstraction, simulation step, control decimation, observation pipeline, and task-space reference generation.
- Explain why the high-fidelity hydraulic model is not executed directly inside every IsaacLab environment.
- Document the stable initial pose, actuator constraints, and relevant simulation settings.

3.6 Hierarchical control architecture

- Reference trajectory to task-space error.
- DLS nominal command and PPO residual action.
- PPO1 interface to ID-FD and PPO2 interface to ROM/AAM.
- Cylinder/valve actuation and feedback to the next control step.

3.7 Problem formulation

- Define the tracking objective and trajectory classes.
- Define model-prediction objectives for ID, FD, and ROM.
- Define robustness and adaptation objectives under parameter variation.
- Define constraints on valve command, cylinder motion, pressure, joint limits, and control smoothness.

3.8 Evaluation criteria introduced at system level

- Task-space error, cylinder-state error, valve error, pressure-related measures, control effort, saturation, stability, and computational cost.
- Explain which measures assess the model and which assess the complete controller.

Chapter 4: Data Generation and Preparation

Explain how the Simscape datasets were designed to contain realistic operation, causal actuator excitation, and variation required for robust and adaptive learning.

Chapter purpose	Main output	Indicative length
Explain how the Simscape datasets were designed to contain realistic operation, causal actuator excitation, and variation required for robust and adaptive learning.	A defensible dataset methodology showing why the data support ID, FD, ROM, latent learning, and AAM training.	10-16 pages

4.1 Dataset objectives and design requirements

- Coverage of the operational workspace and relevant cylinder states.
- Persistent excitation of valve-to-pressure and valve-to-motion dynamics.
- Variation across hydraulic and mechanical parameter families.
- Separation of train, validation, and test systems to evaluate generalization.

4.2 Randomized parameter space

- Describe randomized hydraulic parameters such as effective bulk modulus or air content, valve gain/offset, leakage, and pressure settings.

- Describe randomized mechanical parameters such as mass, center of mass, inertia, stiffness, damping, and payload where applicable.
- Justify ranges physically and distinguish nominal, in-distribution, and out-of-distribution test ranges.
- Explain that randomization supplies variation for both robust training and latent adaptation; it is not merely test noise.

4.3 Closed-loop data collection

- Describe the reference trajectories and controller used in Simscape.
- Explain the benefit of realistic state distributions and stable operating behaviour.
- Discuss the limitation: feedback can correlate the state history and valve command so strongly that the learned dynamics may ignore valve authority.

4.4 Open-loop excitation experiments

- Describe valve excitation signals, amplitudes, switching rates, safety constraints, and initial conditions.
- Explain how open-loop data exposes causal valve-to-pressure and valve-to-cylinder transitions that may be hidden in closed-loop trajectories.
- Discuss coverage of reversals, transients, low-motion regions, saturation, and individual-cylinder excitation.

4.5 Combining closed-loop and open-loop datasets

- Explain the selected mixture or weighting strategy.
- Prevent one data source from dominating training.
- Retain run-level grouping to avoid leakage between train and validation sets.
- Document whether fine-tuning uses the open-loop data after pretraining on the larger closed-loop set.

4.6 Feature and target construction

- Define cylinder-space input variables: valve command, differential pressure, extension, extension velocity, joint state where retained, and optional forces.
- Define ID, FD, and ROM targets and whether they are absolute values, increments, or residuals.
- Define temporal history length, sample interval, causality, and previous-action alignment.

4.7 Preprocessing and data splits

- Filtering, derivative estimation, normalization, outlier handling, and invalid-run rejection.
- Run-level and parameter-level split design.
- Preservation of rare pressure, velocity, and valve regimes.

4.8 Dataset diagnostics

- Workspace and cylinder-range coverage.
- Valve distribution and valve-to-motion sensitivity.
- Pressure and transient coverage.
- Correlation, singular-value, ablation, and persistence-baseline checks for identifiability.
- Final justification of why the combined dataset is suitable for each learned component.

Why the Simscape randomization was necessary

The AAM cannot learn to infer changing dynamics from a dataset containing only one system. Randomization produces multiple plausible systems; the latent representation organizes those variations; the AAM learns to infer the active variation from recent behaviour; and the ROM uses that estimate to adapt its state transition.

Chapter 5: Learned Models and Online Adaptation

Develop the learned actuator representations, explain their training objectives, and establish the distinction between robust modeling and explicit online adaptation.

Chapter purpose	Main output	Indicative length
Develop the learned actuator representations, explain their training objectives, and establish the distinction between robust modeling and explicit online adaptation.	Validated ID, FD, ROM, latent representation, and AAM models with model-level evidence before control integration.	14-22 pages

5.1 Modeling overview and common notation

- Present a single diagram showing the inputs, history windows, outputs, and relationships among ID, FD, ROM, teacher latent representation, and AAM.
- Define shared normalization, causal timing, and cylinder ordering conventions.

5.2 Inverse dynamics model

- State the input and output contract: current/history state plus desired motion to valve command.
- Describe architecture, loss, regularization, command limits, and training procedure.
- Evaluate valve-command error by cylinder and operating regime.

5.3 Forward dynamics model

- State the input and output contract: current/history state and valve input to future cylinder or pressure response.
- Explain one-step versus rollout prediction.
- Show that the model uses valve information rather than only state persistence.
- Clarify that the FD is not conditioned by the AAM in this thesis.

5.4 Reduced-order model

- Define the reduced state and predicted transition.
- Justify why the selected state captures the dominant actuator behaviour needed by PPO2.
- Describe nominal, randomized/robust, and latent-conditioned variants without treating them as unrelated models.

5.5 Latent representation of randomized dynamics

- Explain how the teacher or encoder obtains a compact representation from randomized-system context.
- State the latent dimension and training objective.
- Analyze whether latent clusters or trends correlate with parameter variations, while avoiding unsupported one-latent-one-parameter claims.

5.6 Arm Adaptation Module

- Define the observation history available online.
- Describe the temporal estimator that predicts the teacher latent vector.

- Explain causal deployment, update rate, warm-up history, and behavior after sudden or gradual parameter changes.

5.7 Training and model-selection protocol

- Common train/validation/test splits.
- Hyperparameters, early stopping, checkpoint selection, and random seeds.
- Loss weighting across cylinder extension, velocity, pressure, or other target groups.

5.8 Offline model evaluation

- One-step metrics and multi-step rollout metrics.
- Nominal, randomized seen, and unseen-parameter performance.
- AAM latent-estimation error and adaptation delay.
- Ablations for valve input, history length, latent conditioning, and dataset source.

Chapter 6: Control Framework and Simulation Integration

Explain the complete hierarchical controller and present PPO1 and PPO2 as two controlled experimental pipelines rather than a collection of disconnected models.

Chapter purpose	Main output	Indicative length
Explain the complete hierarchical controller and present PPO1 and PPO2 as two controlled experimental pipelines rather than a collection of disconnected models.	A reproducible controller architecture, PPO task definition, integration details, and fair configuration of all methods.	12-18 pages

6.1 Nominal task-space controller

- Reference-pose generation and task-space error.
- DLS inverse differential kinematics and damping strategy.
- Velocity or position-increment limits and mapping to cylinder targets.
- Role of DLS as the nominal structured controller rather than the adaptive contribution.

6.2 Residual PPO formulation

- Define whether PPO corrects task-space commands, joint/cylinder targets, or low-level commands.
- Observation groups, action scaling, action filtering, and temporal history.
- Reward terms for tracking, smoothness, stability, limits, pressure/effort penalties, and residual magnitude.
- Training curriculum, episode termination, policy network, and PPO hyperparameters.

6.3 PPO1-A: nominal ID-FD pipeline

- ID and FD trained on nominal data.
- No AAM and no latent conditioning.
- Purpose: nominal learned-dynamics baseline under residual PPO.

6.4 PPO1-B: randomized/robust ID-FD pipeline

- ID and FD trained or fine-tuned using randomized and informative data.
- No AAM on the FD.

- Purpose: determine whether data diversity alone improves the learned ID-FD pipeline.

6.5 PPO2-A: nominal ROM pipeline

- ROM trained on nominal data and used with the PPO2 policy architecture.
- No latent conditioning and no AAM.
- Purpose: baseline for the reduced-order architecture.

6.6 PPO2-B: randomized/robust ROM pipeline

- ROM trained across randomized systems but without online latent conditioning.
- Purpose: isolate robustness from adaptation.

6.7 PPO2-C: AAM-conditioned ROM pipeline

- The AAM estimates the current latent dynamics from recent observations.
- The latent estimate conditions the ROM online.
- Purpose: complete proposed adaptive method.

6.8 Optional oracle-latent diagnostic

- Supply the true simulation latent/context to the ROM only as an upper-bound diagnostic.
- Use the gap between oracle and AAM performance to distinguish ROM capacity limitations from adaptation-estimation limitations.

6.9 IsaacLab integration and timing

- Control and physics time steps, action decimation, observation timing, and history-buffer updates.
- Model inference order and handling of previous actions.
- Device placement, vectorized environments, checkpoint loading, and runtime cost.

6.10 Stability and constraint handling

- Joint, cylinder, velocity, pressure, and valve limits.
- Action clipping, rate limits, termination conditions, and numerical safeguards.
- How residual authority is bounded so PPO cannot completely override the nominal controller.

Chapter 7: Experimental Evaluation and Discussion

Test the model and control hypotheses using matched conditions, explain the results rather than only reporting them, and separate robustness from adaptation claims.

Chapter purpose	Main output	Indicative length
Test the model and control hypotheses using matched conditions, explain the results rather than only reporting them, and separate robustness from adaptation claims.	A complete evidence chain from offline model quality to closed-loop tracking and adaptation performance.	18-28 pages

7.1 Evaluation questions and hypotheses

- Restate each research question as a measurable hypothesis.
- Define beforehand what result would support or reject each claim.

7.2 Experimental matrix and fairness controls

- PPO1-A nominal ID-FD.
- PPO1-B randomized/robust ID-FD.
- PPO2-A nominal ROM.
- PPO2-B randomized/robust ROM.
- PPO2-C AAM-conditioned ROM.
- Optional PPO2-D oracle-latent ROM.
- Use matched reference trajectories, parameter sets, simulation step, action scale, policy capacity, training budget, seeds, and evaluation horizon wherever technically possible.

7.3 Model-level experiments

- ID valve prediction under nominal and randomized conditions.
- FD and ROM one-step and rollout prediction.
- Valve-input ablations and persistence baselines.
- AAM latent estimation under fixed and changing systems.

7.4 PPO training behaviour

- Learning curves, reward decomposition, success rate, stability, and seed variance.
- Evidence that improved model quality changes sample efficiency or final performance.

7.5 Task-space trajectory tracking

- Stationary-arm tracking while the vehicle is fixed.
- Vehicle movement to a new station followed by a new tracking task, if included in the final scope.
- Reference shapes such as straight line, circle, arc, and representative free-form curve.
- Report end-effector position/orientation error together with cylinder and control metrics.

7.6 Robustness and generalization tests

- Nominal parameters.
- Seen randomized parameter combinations.
- Unseen combinations within the training envelope.
- Out-of-distribution hydraulic and mechanical variations.
- Payload, disturbance, pressure, leakage, valve, or latency tests according to the final implemented system.

7.7 Adaptation experiments

- Fixed parameter mismatch from the beginning of an episode.
- Abrupt parameter change during operation.
- Gradual drift where feasible.
- Measure adaptation delay, latent error, transient tracking error, and post-adaptation recovery.

7.8 Ablation studies

- Nominal ROM versus randomized ROM isolates data-driven robustness.
- Randomized ROM versus AAM-conditioned ROM isolates explicit online adaptation.
- AAM-conditioned versus oracle-latent ROM isolates adaptation-estimation quality.
- Closed-loop-only versus combined closed/open-loop training isolates the value of informative excitation.
- Optional removal of PPO residual, where safe and meaningful, shows how much correction is supplied by the policy.

7.9 Metrics and statistical reporting

- End-effector RMSE, maximum error, settling or recovery time, and path completion.
- Cylinder extension/velocity error, pressure-related error, valve error, saturation, command variation, and residual-action magnitude.
- Mean, standard deviation, confidence intervals, and per-seed or per-parameter distributions.
- Inference time, memory use, and achievable simulation throughput.

7.10 Integrated discussion

- Explain why each method succeeded or failed in terms of dataset information, model structure, latent estimation, residual authority, and actuator behaviour.
- Separate architecture effects from adaptation effects.
- Discuss whether PPO masks weak model dynamics and use residual magnitude/model rollouts to support the interpretation.
- Relate findings back to the literature and research gap from Chapter 2.

7.11 Threats to validity

- Simscape-to-IsaacLab abstraction mismatch.
- Simulation-to-real limitations.
- Finite randomized parameter ranges and incomplete operating coverage.
- Policy stochasticity, model-identifiability limitations, and measurement assumptions.

Chapter 8: Conclusion and Outlook

Summarize what was established, answer the research questions directly, and identify the next steps without introducing new experimental evidence.

Chapter purpose	Main output	Indicative length
Summarize what was established, answer the research questions directly, and identify the next steps without introducing new experimental evidence.	A concise final statement of contributions, limitations, and future research directions.	4-7 pages

8.1 Summary of the approach

- Summarize the dataset, learned models, control pipelines, adaptation mechanism, and evaluation in the same order as the thesis.

8.2 Answers to the research questions

- Provide one direct evidence-based answer per research question.
- State where the evidence is positive, mixed, or insufficient.

8.3 Main contributions

- Restate only contributions demonstrated by the results.
- Distinguish implemented engineering contributions from scientific findings.

8.4 Limitations

- Model and data limitations.

- Adaptation limitations.
- Simulation and transfer limitations.
- Control and computational limitations.

8.5 Future work

- Real-system data collection and sim-to-real validation.
- Joint training or end-to-end refinement of the ROM and AAM.
- Uncertainty-aware or probabilistic latent adaptation.
- Safety-constrained residual RL and pressure-aware control.
- AAM conditioning of alternative learned dynamics only when scientifically justified by future work.

14. Recommended Appendices

Appendix	Suggested title	Content
Appendix A	System parameters and coordinate mappings	Joint/cylinder ranges, sign conventions, mapping validation, initial pose, and simulation constants.
Appendix B	Dataset and randomization details	Parameter ranges, trajectory-generation settings, run split, preprocessing, and dataset statistics.
Appendix C	Model architectures and training settings	Layer definitions, normalization values, loss weights, optimizers, schedules, and checkpoint selection.
Appendix D	PPO and IsaacLab configuration	Observations, actions, rewards, termination rules, environment settings, and training hyperparameters.
Appendix E	Supplementary results	Per-cylinder metrics, per-seed curves, additional trajectories, failure cases, and extended ablations.

15. Indicative Page Allocation

Chapter	Indicative pages
1. Introduction	6-9
2. Fundamentals and Related Work	12-18
3. System Fundamentals and Problem Formulation	10-15
4. Data Generation and Preparation	10-16
5. Learned Models and Online Adaptation	14-22
6. Control Framework and Simulation Integration	12-18
7. Experimental Evaluation and Discussion	18-28
8. Conclusion and Outlook	4-7
Main-text total	86-133

Important
The page ranges are planning estimates, not targets that should force unnecessary text. The strongest thesis is one in which each chapter contains the evidence required for its purpose and avoids repeating the same explanation in multiple places.

16. Chapter-Writing Checklist

- Begin each chapter with a short paragraph stating its purpose and connection to the previous chapter.
- End each technical chapter with a summary of what was established and what the next chapter builds upon.
- Use one notation consistently across Simscape, offline model training, and IsaacLab.
- Introduce every figure and table in the text before it appears and explain the conclusion the reader should draw from it.
- Keep implementation details that are necessary for reproducibility in the main text; move exhaustive configuration dumps to appendices.
- Do not call a method adaptive merely because it was trained on randomized data; reserve adaptation for the online AAM-conditioned ROM.
- Do not claim that latent dimensions are physical parameters unless this is empirically demonstrated.
- Keep PPO constant across the main ablations so the comparison isolates nominal, robust, and adaptive actuator representations.
- Report failed or negative findings, especially data-identifiability limitations, because they justify the open-loop data and strengthen the methodological contribution.